

Project Development Phase

Model Performance Test (Tableau Analytics Dashboard)

Date	October 03,2026
Team ID	SWTID-2026-7210
Project Name	Script Controlled ACL: Restrict Record Access Based On Field Value
Maximum Marks	10 Marks

Model Performance Testing

Project team shall fill the following information in model performance testing for Tableau evaluation metrics.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	• Data Source Connection: Live connection established to the secure PostgreSQL transactional logging database engine. • Record Mapping Fields: Fetched 450,000 real-time security audit trails, containing active columns: user_id, record_id, access_status, access_timestamp, and dynamic security flags. • Refresh Rate Latency: Hyper-extracted synchronization optimization limits database pipeline response delays comfortably under 250 milliseconds.
2.	Data Preprocessing	• Custom Pivot Layer Transforms: Standardized dynamic multi-value relational strings into individual rows within Tableau's native schema parser structure. • Data Type Modifications: Enforced explicit classification masks setting record security flags into localized Boolean filters (`Is_Confidential` true/false checks). • Handling Null Elements: String cleanings executed converting unassigned null status fields to a secure default baseline flag (`Status = 'Restricted')`).

3.	Utilization of Filters	● Context Filter Optimization: Hardcoded structural baseline filters to process organizational access logs before loading background sheet graphics. ● Interactive Dashboard Slicers: Deployed interactive dynamic drop-down selectors for users to instantly filter security metrics by User Role, Field Label Status, and Audit Failure Codes. ● Row-Level Security Enforcements: Embedded Tableau's `ISMEMBEROF()` user context checks to lock dashboard tracking details based on the viewer's enterprise privilege profile.
4.	Calculation fields Used	● Access Denial Rate % Formula: `COUNTD(IF [Access Status] = 'DENIED' THEN [Transaction ID] END) / COUNTD([Transaction ID])` ● Field Status Context Level Flag Mapping: `IF [Security Field Level] = 'Confidential' AND [User Privilege Level] != 'Admin' THEN 'BLOCKED' ELSE 'ALLOWED' END` ● Mean Interception Latency SLA Tracker Check: `AVG(DATEDIFF('millisecond', [Request Timestamp], [Evaluation Complete Timestamp]))`
5.	Dashboard design	● No of Visualizations / Graphs: 5 Interactive Analytical Sheets Layout Components. ● Visualization Breakdown Summary Matrix: - Real-Time KPI Metric Indicators: Displays Active Requests, Total Denials, and Baseline Violation Flags. - Horizontal Clustered Bar Graphs: Tracking access counts sorted by field values (Confidential vs Open). - Time-Series Scatter Plots: Highlights access authorization checking latency timelines. - Heatmap Distribution Layouts: Highlights security failure concentrations across systems.

6.

Report Design / Story Design

- No of Visualizations / Graphs: 4 Narrative Storyboards (Operational Narrative Flow Mapping Lines).
- Sequential Story Layout Checkpoints:
 - Storyboard 1: High-Level Corporate Compliance Security Compliance Overviews (Summary Metrics).
 - Storyboard 2: Root-Cause Investigation Framework Mapping Fields and ACL Logic Errors.
 - Storyboard 3: Real-Time Performance Latency & System Architecture Load Boundary Checks.
 - Storyboard 4: Post-Audit Optimization Summaries outlining successful mitigation logs.

Appendix: Verification Visuals & Workspace Execution Captions

